

RetailMart Inventory Management Rules & SOPs

Supply Chain & Warehouse Operations | Rev 3.1 | March 2026

1. Purpose

This document establishes the Standard Operating Procedures (SOPs) for inventory control, replenishment, shrinkage management, and forecasting across all RetailMart distribution centres and store locations. Adherence ensures product availability, cost efficiency, and accurate reporting.

2. Inventory Classification (ABC–XYZ Matrix)

Class	Criteria	SKU Share	Revenue Share	Review Frequency
A	High value / fast-moving	10%	70%	Daily
B	Medium value / moderate	20%	20%	Weekly
C	Low value / slow-moving	70%	10%	Monthly
X	Steady demand (low variance)	—	—	Automated
Y	Seasonal / predictable swings	—	—	Weekly
Z	Erratic / unpredictable demand	—	—	Manual

All SKUs are classified quarterly. AZ and CZ items are candidates for discontinuation review.

3. Reorder Point & Safety Stock Formula

3.1 Reorder Point (ROP)

$ROP = (Average\ Daily\ Sales \times Lead\ Time\ in\ Days) + Safety\ Stock$

3.2 Safety Stock Calculation

$Safety\ Stock = Z\text{-score} \times Standard\ Deviation\ of\ Demand \times Square\ Root\ of\ Lead\ Time$. Service level targets by category:

Category	Target Service Level	Z-score	Max Stockout Days/Year
FMCG / Grocery	98%	2.05	7
Electronics	95%	1.65	18
Apparel	92%	1.41	29
Home & Furniture	90%	1.28	36
Stationery	88%	1.17	44

4. Replenishment Process

- Step 1 – System Alert: WMS raises a replenishment alert when stock reaches ROP.
- Step 2 – Validation: Store manager validates alert against current shelf count (cycle count).
- Step 3 – PO Generation: System auto-generates Purchase Order to preferred vendor.
- Step 4 – Vendor Acknowledgement: Vendor confirms within 4 hours for A-class, 24 hrs for others.
- Step 5 – GRN: Goods Receipt Note created upon delivery; quality check completed.
- Step 6 – Putaway: Items moved to designated bin locations within 2 hours of GRN.
- Step 7 – System Update: WMS inventory updated; ROP recalculated if demand pattern changed.

5. Lead Time Standards by Supplier Type

Supplier Category	Expected Lead Time	Escalation Trigger	Penalty
Local / City supplier	1–2 days	> 3 days	2% order value
Regional supplier	3–5 days	> 7 days	3% order value
National supplier	5–7 days	> 10 days	4% order value
Imported / International	15–30 days	> 35 days	Per contract
Emergency replenishment	Same day	Not met	Vendor review

6. Inventory Accuracy & Cycle Counting

6.1 Scheduled Cycle Counts

Class	Count Frequency	Method	Tolerance
A items	Weekly	Full blind count	0%
B items	Monthly	Tag count	0.5%
C items	Quarterly	Spot check	1.0%
Full store audit	Annual	Shutdown count	0%

6.2 Shrinkage Thresholds & Actions

Shrinkage %	Category	Required Action
< 0.5%	Acceptable	No action; log in monthly report
0.5% – 1.0%	Elevated	Root-cause analysis within 5 days
1.0% – 2.0%	High	CCTV review + staff interview
> 2.0%	Critical	Escalate to Regional Head + police report if theft suspected

7. Demand Forecasting Guidelines

RetailMart uses a hybrid forecasting model combining:

- Statistical models: Holt-Winters Exponential Smoothing for seasonal items; ARIMA for trend items.
- ML models: XGBoost-based demand forecasting pipeline updated weekly via Azure ML.
- Human override: Category managers may override system forecast $\pm 20\%$ with documented rationale.
- Festive uplift factors: Pre-loaded multipliers applied 4 weeks before major sale events.

8. Expiry & Perishable Management

FEFO (First Expired First Out) is mandatory for all perishable and near-expiry items. Items within 30 days of expiry trigger an automatic markdown of 15%. Items within 7 days trigger 40% markdown and immediate front-shelf placement. Expired items are to be quarantined and disposed of per the Waste Management SOP.

Document Owner: Supply Chain Director | Approved by: COO | Next Review: Sep 2026